

Unit - 3

Thermodynamics

It is a science of energy transfer and its effect on the physical properties of substances. It deals with energy changes taking place in all, physical and chemical changes.

Thermodynamic System

A predefined space at a given position anywhere in the universe which is studied outside under thermodynamics principles is called a system.

The space outside the system is called surrounding. The imaginary or real line or curve between system and surrounding is called boundary of the system.

- Open: Mass transfer and heat transfer permitted
- Closed: Only heat transfer permitted
- Isolated: No heat or mass transfer permitted

Control System

In case of control system the entry and exit of mass is permitted but the volume is constant. eg. A fluid flowing through a pipe.

We can say that control system is an open system.

Closed System

In case of closed system, mass remain constant but the volume may change.

Thermodynamic Properties

Certain characteristic which describe the physical condition of the system are called Thermodynamic Properties.

- All properties describe together, state of the system.
- All thermodynamic properties are macroscopic in nature.

There are two types of properties

- Intensive

- Extensive

Mass independent

Mass dependent

eg. Pressure, temp, density eg. volume, energy, enthalpy, entropy etc

Extensive property = Intensive property
mass

Thermodynamic Equilibrium

Any system is considered in Thermo-equilibrium if it is under

- Mechanical equilibrium

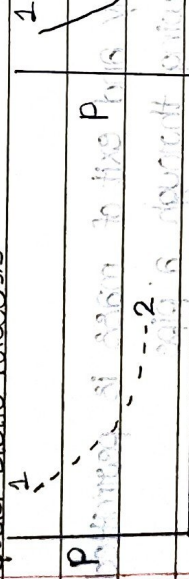
- Chemical equilibrium

- Thermal equilibrium

There is no change in any macroscopic properties when the system is isolated from its surroundings that system will be under

Thermodynamic equilibrium.

Quasi Static Process



Sudden process

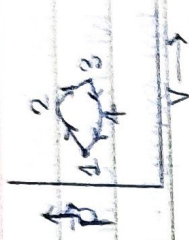
A quasi static process is also called reversible process. This process is a succession of equilibrium state and infinite slowness is its characteristic feature.

Pure substance

Any substance which is homogeneous and invariable in chemical composition throughout its mass. It is to be noted that all considerations are macroscopic.

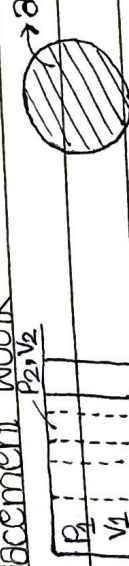
Thermodynamic cycle

A process whose end states are identical

Work Transfer

In case of thermodynamics, the work transfer is the effect on external things which can be represented or replaced solely by lifting a weight. (Not necessarily real, lifting i.e. just an assumption)

The work transfer may be +ve or -ve, work done by the system on surrounding is considered positive and work done on the system by external surroundings is considered negative.

Displacement Work

$$(P_2 - P_1) \cdot dV$$

imperfect/
inexact
differential

$$dW = \text{inexact differential}$$

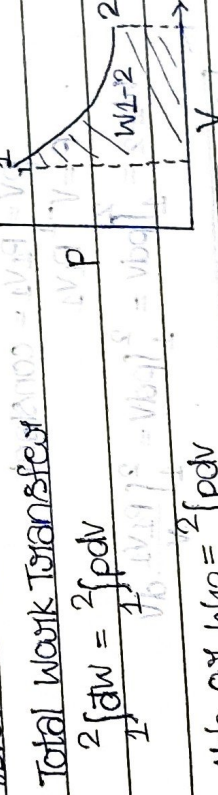
The elemental work transfer due to this small displacement dV

$$dW = F \cdot dV$$

$$= P(dV)$$

$$\boxed{dW = PdV}$$

$$\boxed{dV = \frac{dW}{P}}$$



$$\text{Total Work Transfer}$$

$$\int_1^2 PdV = \int_1^2 PdV$$

$$W_1-2 \text{ or } W_{12} = \int_1^2 PdV$$

$$W_1-2 \neq W_2-1$$

Inexact differential

$$\int_1^2 dV = V_2 - V_1$$

Exact differential

Point function

Those properties in which the change only depends upon final and initial value and is independent of its path. All thermodynamic properties are point function.

Eg. Volume, Enthalpy, Pressure, temperature

Path function

When a change in a property depends upon the path followed also
For example - Heat transfer, Work transfer

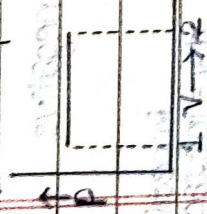
Cyclic Integral of thermodynamic properties

$$\oint dV = 0 \quad \oint dT = 0 \quad \oint dP = 0$$

Cyclic Integral of thermodynamic properties is zero.

Work done for various Thermodynamic processes

1) Isobaric process



$$W_{1-2} = \int_1^2 p dV = p (V_2 - V_1)$$

2) Isochoric process $W_{1-2} = 0$ ($\because dV = 0$)

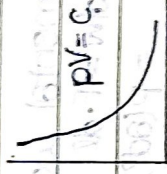
3) Isothermal process ($pV = \text{constant}$)

$$pV = P_1 V_1 = \text{constant}$$

$$P = V^{-1} P_1 V_1$$

$$W_{1-2} = \int_1^2 p dV = \int_1^2 \frac{P_1 V_1}{V} dV$$

$$W_{1-2} = P_1 V_1 \ln \left(\frac{V_2}{V_1} \right) \quad \text{or} \quad W_{1-2} = P_1 V_1 \ln \left(\frac{P_1}{P_2} \right)$$



4) Adiabatic process

$$pV^\gamma = \text{constant}$$

$$W_{1-2} = \frac{P_1 V_1}{\gamma - 1} \left[1 - \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} \right]$$

$$\text{here, } \gamma = \frac{C_p}{C_v} \rightarrow \text{specific heat}$$

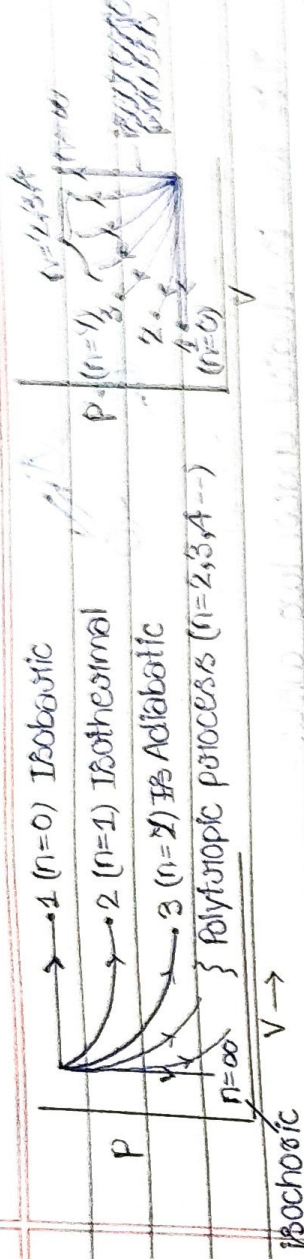
at const. volume



Highest slope than the isotherm for isothermal process

$$[pV^\gamma = \text{constant}]$$

polytropic relation



For Expansion

For Compression

Heat Transfer

Energy exchanged between the system and the surrounding only because of temperature difference. It is a path function.

 dQ is Inexact differential

$$dQ \neq Q_2 - Q_1$$

$$Q_2 - Q_1 = \int_1^2 dQ$$

$$W_2 - W_1 = \int_1^2 PdV$$

$d\alpha = \int_1^2 d\alpha$
 $d\alpha$ - change in entropy of the system
 (extensive property)

$$\int_1^2 d\alpha$$

Specific entropy = $\frac{S}{m}$ (Intensive property)

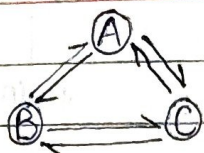
Any heat is added to the system is considered positive and heat going out of the system is considered negative.

Note

- Heat and work are energy in transit
- Both are boundary phenomena
- Both are path functions and inexact differential

Zeroth law of Thermodynamics

If two bodies A and B are in thermal equilibrium, A and C third body are also in thermal equilibrium then B and C will also be in thermal equilibrium. This is one of the most basic laws of nature which was discovered after first law. This law is the basis of various temperature measuring devices such as thermometer, infrared pyrometer, thermocouple.

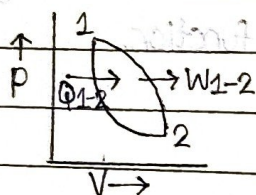


First law of thermodynamics

This law is studied under two cases -

- i) For a closed system undergoing a cycle

When a system undergoes a thermodynamic cycle, then the net heat supplied to the system from the surrounding is equal to the net work done by the system on its surrounding.



$$\oint_1^2 dQ = \oint_1^2 dW$$

cyclic integration

It means that total energy associated with energy conversion remains constant.

PMM1 (Perpetual Motion Machine)

It is an imaginary machine which can provide output in form of work without any energy input.

$$\begin{array}{c} \rightarrow \\ Q=0 \end{array} \boxed{\text{PMM-1}} \rightarrow W_{out}$$

$$\begin{array}{c} \rightarrow \\ Q \\ \text{less input} \end{array} \boxed{\text{PMM-1}} \rightarrow W_{more output}$$

According to first law, PMM-1 is impossible

- ii) For an open process

Heat is considered as a ^{low} grade energy if there is a thermodynamic system in which some energy is added in the form of heat, it provides a work output. Then heat & work are equal and the difference of energy is stored in the system. It is neither heat nor work and is called Internal energy.

$$Q - W = \Delta E$$

where ΔE is a combination of various energies which may

be macroscopic (magnetic, electric) as well as microscopic (molecular or atomic level)

If all the macroscopic energies are negligible then $Q - W = \Delta U$

$$Q = W + \Delta U$$

$$Q = \Delta U + \int_1^2 p dV$$

Enthalpy

It is total amount of heat content which is convertible or non-convertible into work.

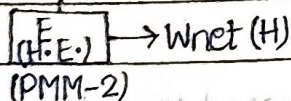
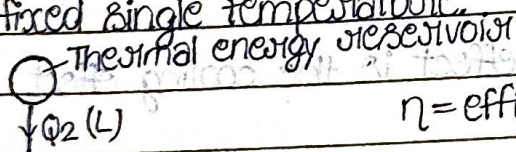
$$H = U + PV$$

If a process absorbs heat entropy change is +ve and if a process releases heat entropy change is -ve

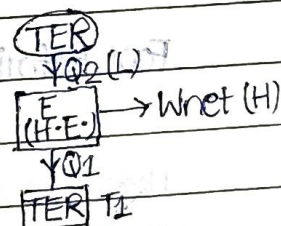
Second law of thermodynamics

It provides information whether a process is feasible or not. Heat is a form of low grade energy and work is a form of high grade energy so, it is not possible that a low level energy is converted completely into a high grade energy.

Kalvin Plank Statement - It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies with a fixed single temperature.



$$\eta = \text{efficiency} = \frac{\text{output}}{\text{input}} \times 100$$

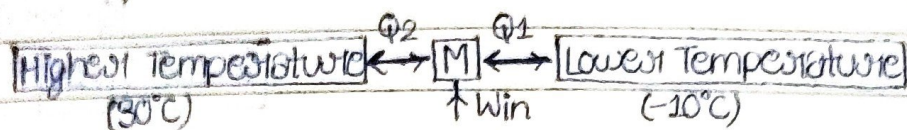


This type of machine is called PMM-2. It is an imaginary machine having 100% efficiency. It is imaginary.

$$\eta_{HE} = \frac{W_{net}}{Q_2} = \frac{Q_2 - Q_1}{Q_2} = 1 - \frac{Q_1}{Q_2}$$

↳ efficiency of heat engine

Clausius Statement - It is impossible to construct a device which operating in a cycle will produce no effect other than the transfer of heat from a cooler object to a ^{hotter} higher object.

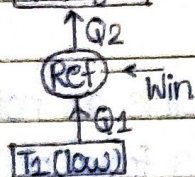


M → Refrigerator

Refrigerator

It is a device operating in a cycle which maintains a body at a temperature lower than the temperature of surrounding.

T_2 (High)



$$[COP] = \frac{\text{Desired effect}}{\text{work input}} = \frac{Q_1}{W_{in}} \quad \text{--- i)}$$

Coefficient of Performance

It is a performance parameter for machines like refrigerator which work on the second law of thermodynamics.

$$[COP] = \frac{\text{Desired effect}}{\text{work input}}$$

For Refrigerators, the desired effect is the cooling effect.

Heat Pump

It is a device which works on the basis of second law of thermodynamics and is used to maintain the temperature of the given space greater than the temperature of surrounding.

$$\therefore (COP)_{HP} = \frac{Q_2}{W_{in}} \quad \text{--- ii)}$$

$$(COP)_{HP} - (COP)_R = \frac{Q_2 - Q_1}{W_{in}}$$

$$[COP]_{HP} = 1 + [COP]_R$$

COP is always greater than 1

Isentropic Process

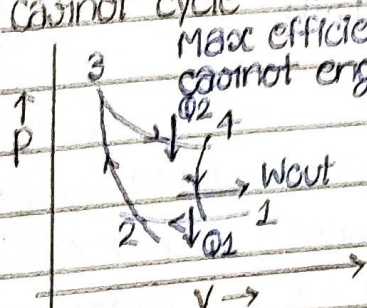
It is the process in which entropy remains constant.

$$S_1 = S_2 \text{ or } S_1 - S_2 = 0$$

$$\int_1^2 \frac{dQ}{T} = \int_1^2 dS \Rightarrow Q = \text{constant}$$

It means that an isentropic process is a reversible adiabatic process.

Carnot cycle



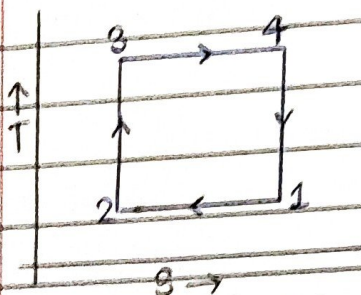
Max efficiency
Carnot engines

- 1-2 Isothermal compression
 - 2-3 Adiabatic compression
 - 3-4 Isothermal expansion
 - 4-1 Adiabatic expansion
- Work producing cycle

Engines based on this cycle are called Carnot engines and such engine may attain

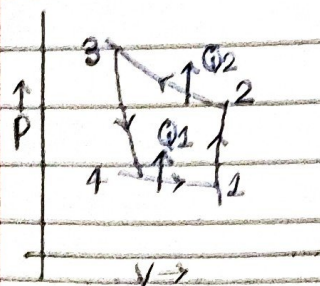
Every process is reversible process.

maximum possible efficiency

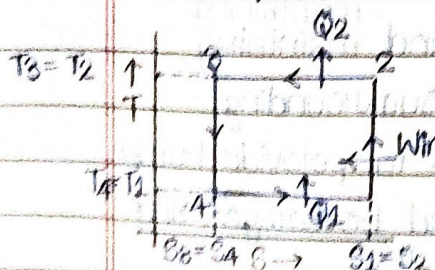


Temperature-Entropy Graph

Reverse Carnot Cycle



- 1-2 → Reversible Adiabatic compression
- 2-3 → Reversible Isothermal compression
- 3-4 → Reversible Adiabatic expansion
- 4-1 → Isothermal expansion



$$Q_1 = \int_4^1 T \cdot dS = T_1 (S_1 - S_4)$$

$$Q_2 = \int_2^3 T \cdot dS = T_2 (S_2 - S_3) = T_2 (S_1 - S_4)$$

$$1 \text{ Pa} = 1.013 \text{ bar}$$

$$1 \text{ bar} = 10^5 \text{ Pascal (N/m}^2\text{)}$$

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$$W_{in} = Q_2 - Q_1$$

$$= T_2 (S_1 - S_4) - T_1 (S_1 - S_4)$$

$$W_{in} = (S_1 - S_4)(T_2 - T_1)$$

Therefore, COP of reverse Carnot refrigerator = $\frac{Q_1}{W_{in}}$

$$= \frac{T_1 (S_1 - S_4)}{(S_1 - S_4)(T_2 - T_1)}$$

$$\Rightarrow (\text{COP})_{\text{Carnot R}} = \frac{T_1}{T_2 - T_1}$$

Similarly, COP of Carnot heat pump = $\frac{Q_2}{W_{in}} = \frac{T_2}{T_2 - T_1}$

$$(\text{COP})_{\text{Carnot HP}} = \frac{T_2}{T_2 - T_1}$$

Ques- In a closed system, 150 kJ of work is supplied if initial volume is 0.6 m^3 and pressure of the system changes as $P = 8 - 4V$ where P is in bars. Determine the final volume and pressure of the system.

$$P = 8 - 4V, W = 150 \text{ kJ}$$

$$V_1 = 0.6 \text{ m}^3$$

$$P \text{ in bars, } V \text{ in m}^3$$

$$V_2 = ?, W = ?$$

$$W = \int_{V_1}^{V_2} P dV$$

$$= \int_{0.6}^{V_2} (8 - 4V) dV$$

$$= [8V - 2V^2]_{0.6}^{V_2}$$

$$= (8V_2 - 2V_2^2) - (8 \times 0.6 - 2 \times (0.6)^2)$$

here, P is given in bars

$$\Rightarrow -150 \times 10^3 = 10^5 (2V_2^2 + 8V_2 - 4.8 + 2 \times (0.6)^2)$$

$$2V_2^2 + 8V_2 - 2.58 = 0$$

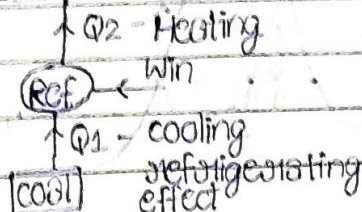
$$V_2 = 4.3 \text{ or } V_2 = -0.3$$

Refrigeration

It is a science which deals with creating and maintaining temperature of a given space below that of surrounding temperature. The purpose of refrigeration is to provide human comfort and preservation of food items and beverages and specific application in some industries.

Refrigerating Effect

It is the amount of heat which is required to be removed from a space to create a cooling effect. Normally, it is measured in kJ/hour, kJ/minute and J/second.



$$Q_2 = Q_1 + W$$

There is one more popular unit which is 1TR (Tonne) of refrigeration

1TR (Tonne) is the amount of heat which is to be removed from 1 tonne of water at 0°C to convert it into 1 ton of ice at 0°C in 24 hours

$$1TR = \frac{mL}{24 \times 60} = \frac{907.15 \times 334 \text{ (kg) (kJ/kg)}}{24 \times 60}$$

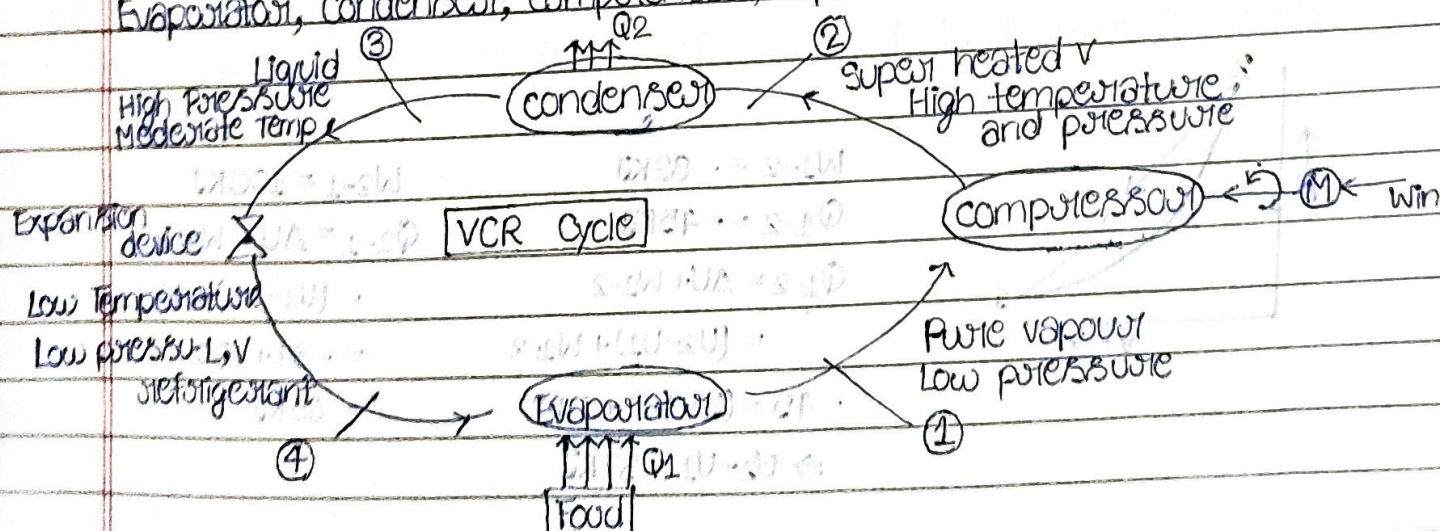
(long ton)
In India, 1000kg = 1 Ton
Globally, 1 Ton = 2000 Pounds (lb)
= 907.15kg

$$1TR = 210 \text{ kJ/min} = 3.5 \text{ kJ/s} = 3.5 \text{ kW}$$

Vapour Compression Refrigeration System (VCRS)

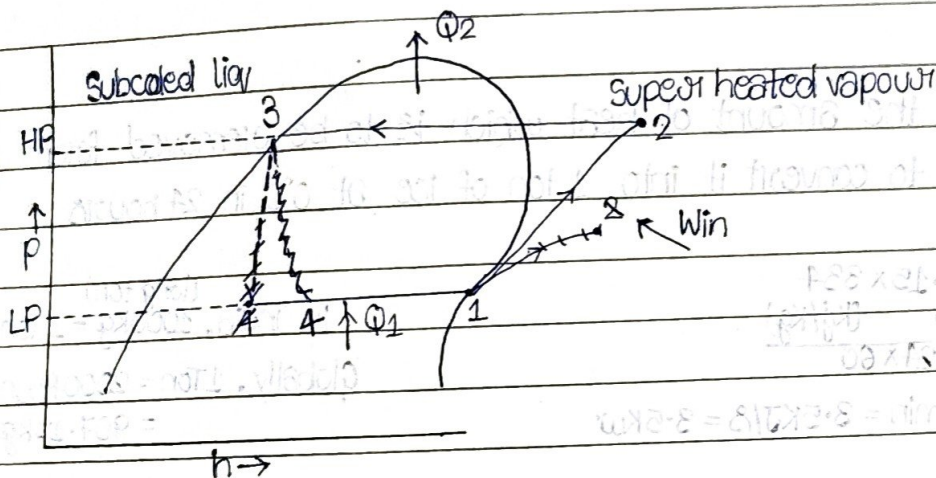
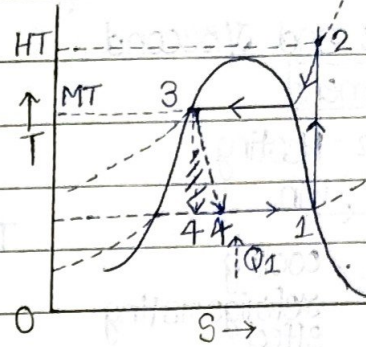
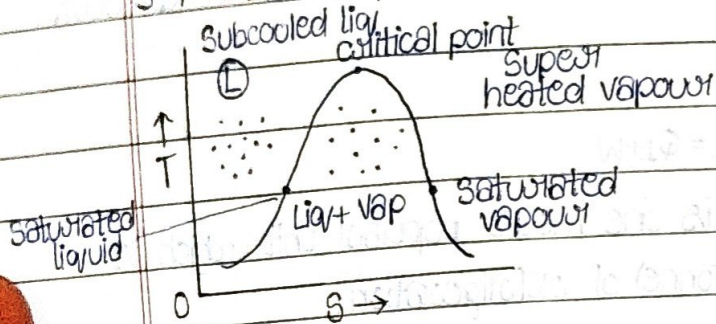
It is an arrangement or system of number of devices working together in a well-defined sequence to give the refrigeration effect. The main components of VCRS are -

Evaporator, Condenser, Compressor, Expansion device



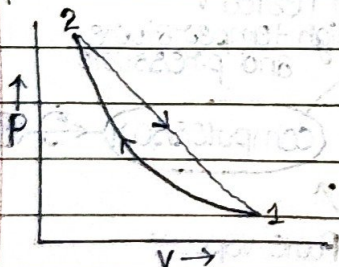
1-2 → Isobaric Heat Addition

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1-2 $\rightarrow$ Reversible Adiabatic (Isentropic) compression2-3 $\rightarrow$ Isobaric Heat Rejection (Q_2)3-4 $\rightarrow$ Isenthalpic expansion ($h_3 = h_4$)

Ques- A cylinder containing the air comprises the system. Cycle is completed as follows: 82000 Joules of work done by the system on the air by the compression stroke and 45 kJ of heat is rejected to the surrounding.

ii) During expansion stroke 100 kJ of work done by the air on the piston. Calculate heat transfer to the system or from the system?



$$W_{1-2} = -82 \text{ kJ}$$

$$W_{2-1} = 100 \text{ kJ}$$

$$Q_{1-2} = -45 \text{ kJ}$$

$$Q_{2-1} = \Delta U + W_{2-1}$$

$$Q_{1-2} = \Delta U + W_{1-2}$$

$$= (U_1 - U_2) + W_{2-1}$$

$$= (U_2 - U_1) + W_{1-2}$$

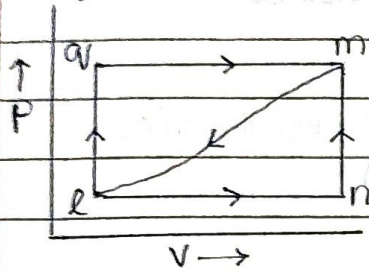
$$= -37 + 100$$

$$-45 = (U_2 - U_1) + (-82)$$

$$= 63 \text{ kJ}$$

$$\Rightarrow U_2 - U_1 = 37 \text{ kJ}$$

Ques- When a system is taken from state L to M along the path LQM 168 kJ of heat does flows into the system and system does 64 kJ of work. How much will be the heat that flows into the system along the path LMN if the work done is 24 kJ?



$$Q_{LQM} = 168 \text{ kJ}$$

$$W_{LQM} = 64 \text{ kJ}$$

$$Q_{LQM} = \Delta U + W_{LQM}$$

$$Q_{LQM} = (U_M - U_L) + W_{LQM}$$

$$168 = (U_M - U_L) + 64$$

$$\Rightarrow [U_M - U_L = 104 \text{ kJ}]$$

$$W_{LMN} = 24 \text{ kJ}$$

$$Q_{LMN} = \Delta U + W_{LMN}$$

$$= (U_M - U_L) + 24$$

$$= 104 + 24$$

$$Q_{LMN} = 128 \text{ kJ}$$

When the system is returned from M to L along the curved path the work done on the system is 42 kJ. Does the system absorb or release heat? How much?

$$W_{ML} = -42 \text{ kJ}$$

$$Q_{ML} = \Delta U + W_{ML}$$

$$= (U_L - U_M) + W_{ML}$$

$$= -104 - 42$$

$$= -146 \text{ kJ}$$

146 kJ of heat is released by the system

If $U_L = 0$ and $U_N = 84 \text{ kJ}$ Find the heat absorbed in the process

Q_{LN} and Q_{NM} ?

$$W_{LNM} = W_{LN} + W_{NM}$$

$$21 = W_{LN} + 0$$

$$\Rightarrow [W_{LN} = 21 \text{ kJ}]$$

$$Q_{LN} = \Delta U + W_{LN}$$

$$Q_{LN} = (U_N - U_L) + W_{LN}$$

$$Q_{LN} = (84 - 0) + 21$$

$$= 105 \text{ kJ}$$

$$Q_{LNM} = Q_{LN} + Q_{NM}$$

$$125 = 105 + Q_{NM}$$

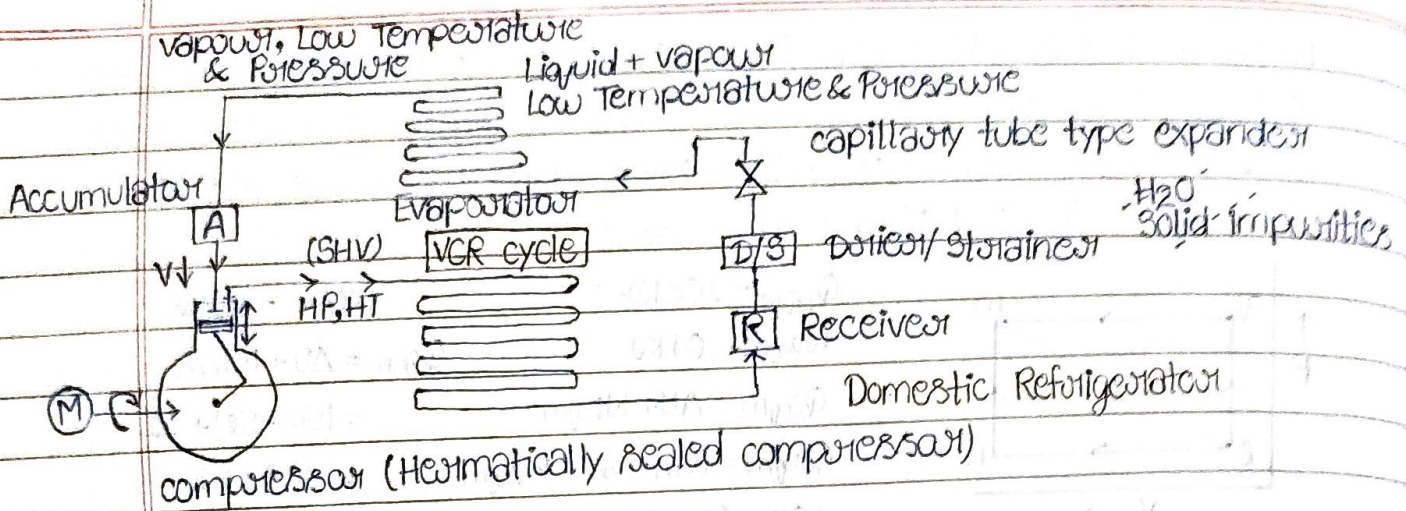
$$\Rightarrow Q_{NM} = 20 \text{ kJ}$$

Functioning of Domestic Refrigerator

$$\frac{Q_2}{Q_1 - Q_2} = 9(100)$$

$$P = \frac{Q_1}{Q_1 - Q_2}$$

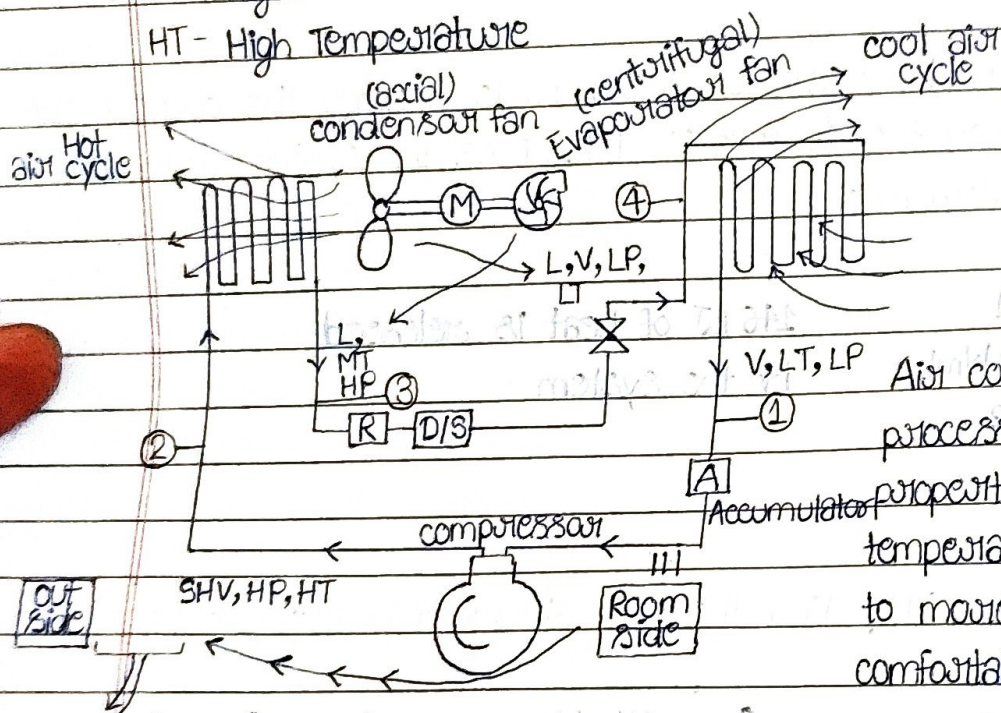
$$= 100$$



SHV - Superheated vapour

HP - High Pressure

HT - High Temperature



Ques-1 A Carnot refrigerator absorbs heat at 270K, and rejects it at 300K. Calculate the COP of this refrigeration cycle. If the cycle is refrigerating absorbing 1130kJ/min at 270K, how many kJ of work is required per second? Also calculate the kJ of heat delivered per minute, at 300K. Also calculate the COP of a heat pump, working between the same temperature?

$1300\text{kJ} = T_2$
 $\uparrow Q_2$
 $\{R\} \leftarrow W_{in}$
 $\uparrow Q_1 \text{ absorbed}$
 $[270\text{K}] = T_1$

$$\begin{aligned}
 (COP)_R &= \frac{270}{300 - 270} \\
 &= \frac{270}{30} = 9
 \end{aligned}$$

Specific heat of water = $4.18 \text{ kJ/kg}^\circ\text{K}$
 latent heat of water = 335 kJ/kg

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$$Q_1 = 1130 \text{ kJ/min}$$

$$(\text{COP})_R = \frac{Q_1}{W_{in}}$$

$$q = \frac{1130}{W_{in}}$$

$$Q_2 = Q_1 + W_{in}$$

$$= 1130 + 125.5$$

$$= 1255.5 \text{ kJ/min}$$

$$\Rightarrow W_{in} = \frac{1130}{q} = 125.5 \text{ kJ/min}$$

$$= \frac{125.5}{60} \text{ kW}$$

$$= 2.09 \text{ kW}$$

$$(\text{COP})_{HP} = 1 + q = 10 = \frac{Q_2}{W_{in}}$$

Ques- 250 litres of drinking water is required per hour at 10°C . Would the use of 1.5 ton refrigerating system be justified if the available water is at 30°C ? If a Carnot engine is made to work between the above mentioned temperature limits what would be its efficiency?

$$m = 250 \text{ kg/hr}$$

30°C water $\rightarrow$ 10°C water

Total Heat to be removed

$$= m s \Delta T$$

$$= 250 \times 4.18 \times 20$$

$$= 20900 \text{ kJ/hr}$$

Capacity of the refrigerator = 1.5 TR

$$= 1.5 \times 12600 \text{ kJ/hr}$$

$$= 18900 \text{ kJ/hr}$$

The Refrigerator is not justified.

$$10^\circ\text{C} = 283\text{K}$$

$$30^\circ\text{C} = 303\text{K}$$

$$\text{Desired Capacity} = \frac{20900}{12600}$$

$$= 1.65 \text{ TR}$$

$$\eta_{\text{Carnot}} = 1 - \frac{T_1}{T_2} = 1 - \frac{283}{303}$$

$$\approx 0.06601$$

Ques- The capacity of a refrigerator is 200 TR when it works between -6°C and 25°C . Determine the mass of ice produced per day from water at 25°C . Also determine the mass of ice power required to drive the unit. Assume that the cycle operates on reversible Carnot and latent heat of ice is 335 kJ/kg . Specific heat of water is $4.187 \text{ kJ/kg}^\circ\text{K}$. (hint: ice is produced at 0°C . So, need not to consider sensible cooling from 0°C to -6°C)

For the above question, determine the mass of ice produced per day, having temperature of -6°C . Specific heat of ice is $2.1 \text{ kJ/kg}^\circ\text{K}$ (hint- here, sensible cooling of ice from 0°C to -6°C is also to be considered)

$$T_1 = 261\text{ K} \quad T_2 = 298\text{ K} \quad \text{Capacity of refrigerator} = 200\text{ TR}$$

$$(\text{COP})_R = \frac{261}{298 - 261} = 6.61 = 200 \times 12600 \text{ KJ/hr}$$

Assuming mass of ice formed/hr = m

Total heat to be removed = Sensible heat + latent heat

$$= m\Delta T + mL$$

$$= m(4.18) \times (25 - 0) + m \times 335$$

$$= 439.5m \text{ KJ/hr}$$

$$439.5m = 200 \times 12600$$

$$\Rightarrow m = 5733.78 \text{ KJ/hr}$$

$$\text{Mass of ice formed/day} = 24 \times 5733.78$$

$$= 137610.9 \text{ kg/day}$$

$$(\text{COP})_R = \frac{Q_1}{W_{in}}$$

$$6.61 = \frac{200 \times 3.5}{W_{in}}$$

$$W_{in} = \frac{200 \times 3.5}{6.61}$$

$$= 81.3 \text{ kW}$$

Let the mass of ice (having temperature -6°C) produced = $y \text{ kg/m}$

Total heat to be removed for ice formation at (-6°C) is

$$= \text{Sensible heat} + \text{heat} \quad \text{heat latent heat} + \text{sensible heat (ice)} = 0^\circ\text{C}$$

$$= m\Delta T_w + m_L + m\Delta T_i$$

$$y \times 4.18 \times 25 + y \times 335 + y \times 2.1 \times 6$$

$$452.1y \text{ KJ/min}$$

$$452.1y = 42000 \Rightarrow y = 92.9 \text{ kg/min}$$

$$x = 133776 \text{ kg/day}$$

Ques- A cold storage is to be maintained maintained at -5°C while the surroundings are at 35°C . The heat leakage from the surrounding into the cold storage is estimated to be 29 kW . The actual COP of the refrigeration plant is one third of the ideal plant working between the same temperatures. Find the power required to drive the plant?

$$[35^\circ\text{C}] = T_2$$

$$Q_1 = 29 \text{ KJ/s}$$

$$(\text{COP})_{\text{ideal}} = \frac{T_1}{T_2 - T_1} = \frac{268}{308 - 268} = 6.7$$

$$[-5^\circ\text{C}] = T_1$$

$$29 \text{ KJ/s}$$

$$\text{Actual COP} = \frac{Q_2}{Q_1} = 2.4/3$$

$$(\text{COP})_{\text{actual}} = \frac{Q_2}{W_{\text{in}}} = \frac{27}{12}$$

$$2.27 = \frac{27}{W_{\text{in}}}$$

$$\Rightarrow W_{\text{in}} = 12 \text{ kJ} \quad (\text{lowest temperature})$$

Ques- 0.2 kg of Nitrogen gas at work and NTC is contained in a cylinder. The piston is moved compressing the gas until the pressure becomes 1 MPa and temperature 20°C. The work done during the process is 30 kJ. Calculate the heat transferred from the to the surrounding. Take C_v of $N_2 = 0.15 \text{ kJ/kgK}$

$$\Delta U = m C_v \Delta T \quad \Delta H = m C_p \Delta T$$

specific heat at constant volume specific heat at constant pressure

$$Q = W + \Delta U$$

$$W = -30 \text{ kJ}$$

$$Q = -30 + m C_v \Delta T$$

$$= -30 + 0.2 \times 0.15 \times 120$$

$$= -3 \text{ kJ}$$

↳ Heat released

specific volume

Ques- Air enters a compressor at 10⁵ Pa and 20°C having volume of 200 kg is compressed to 5 × 10⁵ Pa isothermally. Determine work done i) change in internal energy ii) heat transferred and its value. Work done in isothermal process, $W = P_1 V_1 \ln \left(\frac{P_2}{P_1} \right)$

$$10^5 \times 2.0 \times \ln \left(\frac{5 \times 10^5}{10^5} \right)$$

$$= 2.89 \times 10^5 \text{ J}$$

$$= 289 \text{ kJ}$$

$$\Delta U (\text{change in internal energy}) = \Delta H = 0$$

Because this is an isothermal process (no change in temperature)

$$Q = W + \Delta U$$

$$= -289 \text{ kJ}$$

Date ____/____/____

Ques- A refrigeration system operates on a reversed Carnot cycle. The highest temperature is 35°C and the lowest temperature is -15°C . The capacity is to be 12 tonnes. Determine: a) COP b) Heat rejected from the system per hour c) Power required.

$$(\text{COP})_R = \frac{T_1}{T_2 - T_1} = \frac{258}{308 - 258} = 5.16$$

$$Q_1 = 12 \text{ TR} = 12 \times 12600 \text{ kJ/hr} = 151200 \text{ kJ/hr}$$

$$(\text{COP})_R = \frac{Q_1}{W_{\text{in}}} \quad W_{\text{in}} = \frac{Q_1}{(\text{COP})_R} = \frac{151200}{5.16} = 29302.32 \text{ kJ/hr}$$

$$Q_1 = \text{RE}$$

capacity of refrigerator

$$12 \times 12600$$

$$Q_2 = Q_1 + W_{\text{in}} \quad (1^{\text{st}} \text{ law of Thermodynamics})$$

$$151200 + 29302$$

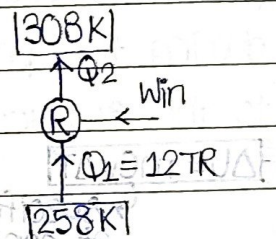
$$180502 \text{ kJ/hr}$$

$$\text{Power required} = W_{\text{in}}$$

$$29302.32 \text{ kJ/hr}$$

$$8.14 \text{ kJ/sec}$$

$$8.14 \text{ kW}$$



$$U \Delta + W = 0$$

$$U \Delta + W = 0$$

$$U \Delta + W = 0$$

$$U \Delta + W = 0$$

$$U \Delta + W = 0$$

$$U \Delta + W = 0$$